

Sixth Semester
Course No.: BCA/6/EC/31
Name of Paper: Elective-II (*any one*)
a) Software Engineering-II

Marks Scale: **100** marks (End Sem. Exam: 75 + Int.: 25)

Credit: **5**

(4- 1- 0)

(10L)

Unit 1:

Software Project Planning: Introduction, Management spectrum 4P (people, process, product, project), Task set of project planning, choosing the right process: Waterfall, Prototype, Iterative Enhancement model, Rapid development Model, Unified process and Spiral models. Agility, Agile Process, Agile Process Models

Unit 2:

(10 L)

Object Oriented Software Engineering: Object Oriented concepts, Object Oriented Analysis: Domain analysis, OOA process, Object-relationship model, Object-behavior model, **Object Oriented Design:** OO design pyramid, translation of OOA to OOD, System design process, Object design process.

Unit 3:

(10 L)

Project Scheduling: Basic Concepts, Project Scheduling: Introduction, Tasks network, Critical Path Method, PERT, Timeline Chart, staffing, Earned Value Analysis, Cost Estimation, COCOMO (basic, intermediate and detailed) and COCOMO II Model.

Unit 4:

(10L)

Software Quality management: Quality concept, SQA: Background and SQA activities, Software reviews and Formal Technical Reviews, Formal approaches to SQA, **Software quality model:** McCall Software Quality Model, FURPS, Boehm Software Quality Model, ISO 9126, CMM, ISO 9000.

Unit 5:

(10L)

UML: Overview of UML, UML diagrams: Class diagram, Object diagram, Use case diagram, Sequence diagram, Collaboration diagram, State machine diagram, Activity diagram, Component diagram, Deployment diagram.

Recommended Books:

1. Pressman Roger S., *Software Engineering- A Practitioner's Approach*, 7th Edition, Tata McGraw Hill-2009
2. Aggarwal KK and Singh Yogesh, *Software Engineering*, 3rd Edition, New Age International Publisher-2008.
3. Pankaj Jalote, *An Integrated Approach to Software Engineering*, 3rd Edition, Narosa Publications, 2011.
4. Jibitesh Mishra, Ashok Mohanty, *Software Engineering*, 1st Edition, Pearson Education, 2011.
5. Ali Bahrami, *Object Oriented System Development*, McGraw Hill International Edition, 2008.

Sixth Semester
Course No.: BCA/6/EC/31
Name of Paper: Elective-II (*any one*)
b) Software Project Management

Marks Scale: **100** marks (End Sem. Exam: 75 + Int.: 25)

Credit: **5**
(4- 1- 0)
10L

Unit-1:

Introduction to Software Project Management:

Introduction, Typical project life cycle, Stepwise: An overview of project planning.
Project Management Concept: People, process, project and product (4P), Project Scheduling, **Project Management Tools:** Gantt Chart, PERT Chart, Software Project Plan Document, System development life cycle (SDLC).

Unit-2:

10L

Project teams, project monitoring and controls –
Formal Technical Review (FTR), Cost estimation – COCOMO Model, Software Metrics – Size oriented and function oriented.
Quality Management - Quality definition, quality management activity, process and product standards, documentation standards, ISO 9126 and CMM, Risk Management, Software Configuration Management (SCM).

Unit-3:

10L

Software effort estimation: Introduction, Software effort estimation techniques, The top-down approach and parametric models, Expert judgement.
Quality models: MC Call's Quality Model, Boehm's Quality Model; **Activity planning:** Objectives, project schedules; **Project and activities:** Activity based approach, The product based approach, The hybrid approach, **Software measurement & tacking planning:** Concept in measurement, Metrics and Measurement, Process monitoring through Statistical Process Control (SPC), Life cycle of defect.

Unit-4:

10L

Resource allocation: The nature of resources, Identifying resource requirements, Scheduling resources, Creating critical paths, Counting the cost, Publishing the resource schedule.
Monitoring & Control: Project control cycle, Collecting the data, Visualizing progress, Cost monitoring, Prioritizing monitoring, Change control.
Managing Contracts: Types of contract, Stages in contract placement, Contract management.

Unit-5:

10L

Managing people and organizing teams: Understanding behavior, Organizing behavior, Selecting the right person for the job, Motivation, Working in groups, Becoming a team, Decision making, Leadership, Organizational structures.
Software quality: Importance of software quality, Software quality measures, Product versus process quality management, Techniques to help enhance software quality.

Recommended Books:

1. Bob Hughes and Mile Cotterell: Software Project Management, 3rd Edn, Tata McGraw Hill (2004)
2. Hughes: Software Project Management, McGraw-Hill Education, 5th Revised edition (2009)

Sixth Semester
Course No.: BCA/6/EC/31
Name of Paper: Elective-II (*any one*)
c) Management Information System

Marks Scale: **100** marks (End Sem. Exam: 75 + Int.: 25)

Credit: **5**
(4- 1- 0)

Unit-1:

System concept, Data and Information, Types of information, Types of Information System at Management levels, **MIS**: Definition, Meaning and Role, System approach, The system view of business, MIS organization within the company, Management Organizational theory and the system approach, Development of organizational theory, Management and organizational behavior, Management, Information and the System approach.

Unit-2:

Data Processing - Operation of Manual Information System, Components of Computer System, Conversion of Manual to Computer Based Systems, Data Bank Concept, types of computer based applications.

Evolution of an Information system, Decision making and MIS, MIS as a technique for making programmed decisions, Decision assisting information.

Unit-3:

Strategic and Project Planning for MIS

General business planning, MIS response, MIS Planning (general & detail).

Define the problem, Set system objectives, Establish system constraints, Determining information needs and sources, Develop alternative conceptual design and select one, Document the system concept, Prepare the conceptual design report.

Unit-4:

Inform and involve the organization, Aim of detailed design, Project management of MIS detail design, Identifying the domain and tradeoff criteria, Defining subsystems, Sketch the detailed operating sub system and information flows, Determination of the degree of automation of each operation, Inform and involve the organization again, Input/Output and processing early system testing, Software, hardware & tools propose to an organization to operate the system, Documentation of the detailed design.

Unit-5:

Planning for the implementation, Acquiring floor space & planning space layout, Organize for the implementation, Develop procedure from implementation, Operating personal training, Computer related acquisitions, Developing forms for data collection and information, Dissemination, Developing the files, Testing the system, Cutover, Documentation of the system, Evaluation of the MIS control and the system maintenance.

Pitfalls in MIS development: Fundamental weaknesses, Soft spot planning, Design problems

Recommended Books:

1. Robert G. Murdick, Joel E. Ross and James R. Claggett: *Information Systems for Modern Management*, PHI
2. Gordan B. Devis: *Management Information System*
3. Jerome Kanter: *Managing with Information*, PHI
4. Kenneth C. Laudon: *Management Information System*, PHI

Sixth Semester
Course No.: BCA/6/EC/32
Name of Paper: Elective III (*any one*)
a) Data Warehousing

Marks Scale: **100** marks (End Sem. Exam: 75 + Int.: 25)

Credit: **5**
(4- 1- 0)

Unit-1:

The Compelling Need For Data Warehousing: Escalating need for strategic information, Failures of past Decision-Support, Operational versus Decision-Support systems, Data Warehousing-The only viable solution, Data Warehouse defined, The Data Warehousing movement, Evolution of Business Intelligence.

Data Warehouse- The Building Blocks: Defining features, Data Warehouses and Data Marts, Architectural types, Overview of the components, Meta data in the Data Warehouse.

Unit-2:

Trends In Data Warehousing: Continued growth in Data Warehousing, Significant trends, Emergence of standards, Web-enabled Data Warehouse.

Defining The Business Requirements: Dimensional Analysis, Information packages, Requirements gathering methods, **Requirement definition:** Scope and content.

Unit-3:

Requirements as The Driving Force for Data Warehousing: Data design, The architectural Plan, Data storage specifications, Information delivery strategy.

Architectural Components: Understanding Data Warehouse Architecture, Architectural framework, Technical architecture, Architectural types.

Unit-4:

Infrastructure as The Foundation For Data Warehousing: Infrastructure supporting architecture, Hardware and Operating Systems, Database software, Collection of tools, Data Warehouse appliances.

The Significant Role of Metadata: Why Metadata is important, Metadata types by function areas, Business Metadata, Technical Metadata, How to provide Metadata.

Unit-5:

Principles of Dimensional Modeling: From requirements to Metadata, The Star schema, Star Schema keys, Advantages of the Star Schema, Star Schema: Examples

Data Mining Basics: What is Data Mining, Data Mining applications.

Recommended Books:

1. Paul Raj Poonia: *Data Warehousing Fundamentals for IT professionals, 2nd Edition*, John Wiley & Sons (2010).
2. Sam Anahony: *Data Warehousing in the Real World: A practical guide for building decisionsupport systems*, John Wiley(2004).
3. W. H. Inmon :*Building the data warehouse*, 4th Ed., John Wiley (2008).
4. Bob :*Data Warehousing*, BPB Publications (2004).

Sixth Semester
Course No.: BCA/6/EC/32
Name of Paper: Elective III (any one)
b) Computer Networking –II

Marks Scale: **100** marks (End Sem. Exam: 75 + Int.: 25)

Credit: **5**

(4- 1- 0)

Unit-1:

(10 L)

Introduction: Attacks (Virus, Worms, Trojan Horses, Hoaxes, Dictionary attack, Brute Force, DOS, DDOS, Phishing, Spoofing, Man-in-the-middle, Mail bombing, Sniffers, Social Engineering), Services and Mechanism, Model for Internetwork Security.

Unit-2:

(10 L)

Cryptography: Plain Text, Ciphertext, Key, Symmetric key cryptography (Traditional ciphers, DES, AES), Asymmetric-Key Cryptography (RSA, Diffie-Hellman), Message Integrity, Message Authentication, Digital Signature, Key management.

Unit-3:

(10 L)

Network Security: Confidentiality, Integrity (Message Digest, SHA-1), Authentication (MAC, HMAC), Digital signature, Entity authentication.

Internet Security: Network Layer Security, Transport Layer Security, Application Layer Security.

Unit-4:

(10 L)

IP Security: AH, ESP, services, Security association, IKE, VPN.

SSL/TLS: SSL (Services, security parameters, sessions and connections, protocols, TLS.

PGP (Security parameters, services, Key rings, Certificates). Firewalls (Packet-filter and Proxy firewalls).

Unit-5:

(10 L)

Security technology: Intrusion detection and prevention system (IDPS), Port scanners, Firewall analysis tools, Operating system detection tools, Vulnerability scanners, Packet sniffers, Wireless security tools. Antivirus Software, Malicious virus remover, worm remover.

Recommended Books:

1. Behrouz A Forouzan: *Data Communication and Networking*, 4th Edition Tata McGraw Hill Publication (2003).
2. Andrew S. Tanenbaum: *Computer Networks*, 5th Edition, Prentice Hall Publication (2010).
3. Michael E. Whitman and Herbert J. Mattort: *Principles of Information Security*, 4th Edition, Cengage Learning (2011).
3. W. Stallings: *Networks Security Essentials: Application & Standards*, Pearson Education (2000).
4. W. Stallings: *Cryptography and Network Security, Principles and Practice*, Pearson Education (2000).

Sixth Semester
Course No.: BCA/6/EC/32
Name of Paper: Elective III (any one)
c) Mobile Computing

Marks Scale: **100** marks (End Sem. Exam: 75 + Int.: 25)

Credit: **5**
(4- 1- 0)

Unit-1:

Introduction

Mobile Computing – Mobile Computing Vs wireless Networking – Mobile Computing Applications – Characteristics of Mobile computing – Structure of Mobile Computing Application. MAC Protocols – Wireless MAC Issues – Fixed Assignment Schemes – Random Assignment Schemes – Reservation Based Schemes.

Unit-2:

Mobile Internet Protocol and Transport Layer

Overview of Mobile IP – Features of Mobile IP – Key Mechanism in Mobile IP – route Optimization. Overview of TCP/IP – Architecture of TCP/IP- Adaptation of TCP Window – Improvement in TCP Performance.

Unit-3:

Mobile Telecommunication System

Global System for Mobile Communication (GSM) – General Packet Radio Service (GPRS) – Universal Mobile Telecommunication System (UMTS).

Unit-4:

Mobile Ad-Hoc Networks

Ad-Hoc Basic Concepts – Characteristics – Applications – Design Issues – Routing – Essential of Traditional Routing Protocols – Popular Routing Protocols – Vehicular Ad Hoc networks (VANET) – MANET Vs VANET – Security.

Unit-5:

Mobile Platforms and Applications

Mobile Device Operating Systems – Special Constrains & Requirements – Commercial Mobile Operating Systems – Software Development Kit: iOS, Android, BlackBerry, Windows Phone – MCommerce – Structure – Pros & Cons – Mobile Payment System – Security Issues.

Recommended Books:

1. Jochen Schiller: *Mobile Communications* Second Edition – Pearson Education
2. William C.Y.Lee: *Mobile Cellular Telecommunications* Second Edition-Mc-Graw-Hill
3. Sandeep Singhal: *The Wireless Application Protocol, Writing Applications for Mobile Internet*, Pearson Education, 2000

Sixth Semester
Course No.: BCA/6/CC/33
Name of Paper: Major Project

Marks Scale: 300 marks (End Sem. Exam: 200 + Int.: 100)

Credit: **14**
(0- 0- 14)

Objective:

The Project work constitutes a major component in most professional program. It needs to be carried out with due care, and should be executed with seriousness by the students. The project work is not only a partial fulfillment of the BCA requirements, but also provide a mechanism to demonstrate your skills, abilities and specialization. The objectives of the project is to help the student develop the ability to apply theoretical and practical tools/techniques to solve real life problems related to industry, academic institutions and research laboratories. The purpose behind the inclusion of this is to consolidate the concepts and practices that were imparted during the course and to serve as record of competence. It should enable the students to apply concretely in a small package the concepts pinned from system analysis and design.

Students should take this project work **very seriously**. Topics selected should be complex and large enough to justify as a BCA project. Please do not undertake the topics/specifications from the Mini Project. The project should be genuine and original in nature and should not be copied from anywhere else.

Mark Distribution:

Marks distribution for internal as well as external is given below:

Topics	External
Project Record	25 Marks
Project Development and execution	100 Marks
Viva Voce	50 Marks
Presentation	25 Marks
Total	200 Marks

Internal – 100 marks

Steps Involved in The Project Work:

The complete project work should be done by the student only. The role of guide should be about guidance wherever any problem encounters during project. The following are the major steps involved in the project, which may help you to determine the milestones and regulate the scheduling of the project:

- 1) Select a topic and a suitable guide.
- 2) Prepare the project proposal in consultation with the project guide.
- 3) Submit the project proposal along with the necessary documents
- 4) Obtain receipt of the project approval from concerned department
- 5) Carry out the project-work.

- 6) Prepare the project report.
- 7) Submit the project report to the department concerned
- 8) Appear for the viva-voce or presentation as per the intimation.

Project Synopsys:

The student can formulate a project synopsis with the help of her/his Guide and submit the project proposal of the same. **Approval of the project proposal is mandatory.** If approved, the student can commence working on it, and complete it. Use the latest versions of the software packages for the development of the project. Synopsis of the project proposal covering the following aspects must be submit to the department:

- (i) Proposed Title of the Project.
- (ii) Why is the particular topic chosen?
- (iii) Introduction and Objectives of the Project.
- (iv) Software Requirement specifications.
 - ❖ Hardware Requirements
 - ❖ Software Requirements
- (v) A complete structure which includes:
 - Number of modules and their description
 - Data Structures as per the project requirements for all the modules
- (vi) List of reports that are likely to be generated

The project synopsis must have proper cover page and signature of students and the guide to show his or her acceptance and willingness to guide.

Acceptance/Rejection of Report:

The student must submit a Synopsis of the project report to the Institute for approval. The Principal/Teacher in charge holds the right to accept the project or suggest modifications for resubmission. Only on acceptance of draft project report, the student should make the final copies.

Project File:

Model 1

1. The topic for the project can be any sub-system of a system software or tool or any scientific or a fairly complex algorithmic situation.
2. The aim of this type is to highlight the abilities of algorithmic formulation, program and data flow representation, modular programming, optimized code preparation and systematic documentation and other associated aspects of software engineering.

Model 2:

1. This model can be of a typical business oriented application. The aim of this type is to highlight the stages involved in a typical business oriented project development, though on a miniature scale and simulated environment. The appropriate use of DBMS/RDBMS towards any business application, along with adequate level system analysis and structured design and development of specific tools/products would be the underlying activity, in a preparing this project.

2. The emphasis should be on selecting a system/ sub- system which shows the DBMS and System Analysis aspects, to a greater degree. Any small and simple business may be selected, although candidate's are advised to use their knowledge and creativity, to select typical and intelligent applications, rather than run-of-the- mill themes, such as simple Pay Roll calculation or Issue Return portion on an inventory scheme. The Evaluation stage would give due weightage for theme selection, problem analysis, fact finding techniques and initial design, which is as close to real life business situations as possible.
3. The project may be earned out in any of the X base family products which are equivalent to Oracle/Access. The code can be generated out of four GL Interface, like Screen Builder and Report Generator, or can be totally hand- coded or a combination of both. The documentation need not contain the code generated by these applications, but only written by the candidate.

Format for Submission of The Project Report:

The student must adhere strictly to the following format for the submission of the Project Report.

Paper:

The Report shall be typed on White paper, A4 size or continuous computer stationery bond, for the final submission. The Report to be submitted to the MizoramUniversity must be original and subsequent copies may be photocopied on any paper.

Typing:

The typing shall be of standard letter size, double spaces and on one side of the paper only, using black ribbons and black carbons.

Margins:

The typing must be done in the following margins:

Left ..35mm, Right ..20mm,

Top ..35mm, Bottom ..20mm.

Binding:

The report shall be rexin bound in black. Plastic and spiral bound Project Reports not is accepted.

Front Cover:

The front cover should contain the following details:

TOP: The title in block capitals of 6mm to 15mm letters.

CENTER: Full name in block capitals of 6mm to 10mm letters.

BOTTOM: Name of the University, Year of submission-all in block capitals of 6mm to 10mm letters separate lines with proper spacing and spacing.

Blank Sheets:

At the beginning and end of the report, two white black bound papers should be provided, one for the purpose of binding and other to be left blank.

Abstracts:

Every report should have an Abstract following the Institute's Certificate. The abstract shall guide the reader by highlighting the important material contained in the individual chapters. The abstract should not exceed 800 words.

Contents:

The contents shall follow the abstract indicating the title of the chapters, section, subsection etc.

Format of The Project Report:

The format of the report should be prepared according to the format given below:

- Cover Page
- Institute Certificate
- Certificate from Company acknowledgements.
- Abstract
- List of Figures.
- List of Tables
- Nomenclature and Abbreviations.
- Contents
- **Introduction**
 - Introduction
 - Objectives
 - Scope and Purpose
- **System Analysis**
 - Need for system
 - Feasibility Study
 - Hardware Requirements
 - Software Requirements Specifications
 - Software Engineering Paradigm applied
 - System Flow Charts
 - Entity Relationship Diagram
 - Data Flow Diagram
- **System Design**
 - Program Structure
 - Modularization Details
 - User Interface Design
 - Menu Explanation
 - Design of Input Output screens and reports
 - Database Design
 - Schema Design & Normalization
 - Data Integrity and Constraints
 - Data Dictionary
- **Coding**
 - Complete Project Coding
 - Comments and Description of Coding segments
 - Standardization of the coding
 - Error Handling Mechanism
 - Parameters calling and passing
 - Validation checks
- **Testing**
 - Testing techniques and Testing strategies
 - Testing Case Design
 - Test reports
- **Drawbacks and Limitations**

- Conclusion
- Bibliography

Viva-Voce and Presentation:

The viva-voce will be conducted by an external examiner appointed by the University and an internal examiner from the College. Other members of the faculty and students may be present. It will be of a duration of about 15 to 20 minutes. The logic, analysis and design aspects relevant to the project mentioned under assessment would be the main subject matter for the viva. However, the general proficiency of the candidate in the selected software platform should also be tested.

Number of Copies:

The student should submit four (4) hard-bound copies of the Project Report and Soft copy on CD/DVD to the college or university which will be forward as given below:

- (1) One copy to their college
- (2) One copy to the university or Board
- (3) One copy for the project guide
- (4) One copy for student which they may collect after getting all necessary signatures.
